

Initial Project Planning Template

Date	15 March 2026
Team ID	
Project Name	Credit Card Approval Prediction using Machine Learning
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Understanding	USN-1	As a data scientist, I can collect historical credit card applicant data so that I have a reliable dataset to train the model.	2	High	Zaamir Hanan Mujeeb	15 Mar 2026	21 Mar 2026
		USN-2	As a data scientist, I can perform exploratory data analysis to understand feature distributions, class balance, and missing values.	2	High	Rushikesh Kasthuri	15 Mar 2026	21 Mar 2026

	Data Preprocessing & Feature Engineering	USN-3	As a data scientist, I can handle missing values and outliers so that the dataset is clean for model training.	3	High	Deepak Bellam	15 Mar 2026	21 Mar 2026
		USN-4	As a data scientist, I can convert multi-class payment status codes into binary risk labels so that high-risk applicants can be clearly classified.	2	High	Mohana Krishna Reddy	15 Mar 2026	21 Mar 2026
		USN-5	As a data scientist, I can encode categorical variables and scale numerical features so that the data is ready for model input.	2	Medium	Shaik Asliman Thaslimaan	15 Mar 2026	21 Mar 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-2	Model Building	USN-6	As a data scientist, I can train a Logistic Regression model so that I have a baseline approval prediction model.	3	High	Zaamir Hanan Mujeeb	22 Mar 2026	28 Mar 2026
		USN-7	As a data scientist, I can train a Decision Tree classifier so that I can compare its performance against other models.	2	Medium	Rushikesh Kasthuri	22 Mar 2026	28 Mar 2026
		USN-8	As a data scientist, I can train a Random Forest classifier so that I can improve prediction accuracy using ensemble learning.	3	High	Deepak Bellam	22 Mar 2026	28 Mar 2026
		USN-9	As a data scientist, I can train an XGBoost (Gradient Boosting) model so that I can achieve higher predictive performance.	3	High	Mohana Krishna Reddy	22 Mar 2026	28 Mar 2026
	Model Evaluation & Selection	USN-10	As a data scientist, I can evaluate all four models using accuracy, precision, recall, and F1-score so that I can compare their performance.	2	High	Shaik Asliman Thaslimaan	22 Mar 2026	28 Mar 2026

		USN-11	As a data scientist, I can save the best-performing model as a serialized file so that it can be integrated into the web application.	1	High	Zaamir Hanan Mujeeb	22 Mar 2026	28 Mar 2026
Sprint-3	Web Application Development	USN-12	As a user, I can enter applicant details (income, employment, credit history) into a web form so that I can request a prediction.	3	High	Rushikesh Kasthuri	29 Mar 2026	04 Apr 2026
		USN-13	As a user, I can view the predicted approval or rejection result on the web page so that I know the outcome instantly.	2	High	Deepak Bellam	29 Mar 2026	04 Apr 2026
		USN-14	As a developer, I can integrate the trained ML model with the Flask backend so that predictions are served through the web app.	3	High	Mohana Krishna Reddy	29 Mar 2026	04 Apr 2026
	High-Risk Applicant Screening	USN-15	As a compliance officer, I can batch-screen applicants with past-due loan records so that high-risk applicants are flagged as ineligible.	3	Medium	Shaik Asliman Thaslimaan	29 Mar 2026	04 Apr 2026
Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-4	Cloud Deployment	USN-16	As a developer, I can deploy the trained model to IBM Watson Machine Learning so that predictions can be served at scale from the cloud.	3	Medium	Zaamir Hanan Mujeeb	05 Apr 2026	11 Apr 2026
		USN-17	As a developer, I can connect the Flask application to the IBM Watson deployment endpoint so that the UI uses the cloud-hosted model.	2	Medium	Rushikesh Kasthuri	05 Apr 2026	11 Apr 2026

	Testing & Validation	USN-18	As a QA engineer, I can test the web application end-to-end so that predictions are accurate and the UI works as expected.	2	High	Deepak Bellam	05 Apr 2026	11 Apr 2026
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Note: Team ID is still a placeholder — please replace it with your actual team ID before submission. Team Member assignments above are distributed across the backlog and can be adjusted to match your actual sprint planning.